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BSCS 4B

Lab 1: Image Manipulation

In this activity, I learned that digital images in computer vision are represented as 2D or 3D arrays of pixel values. Using OpenCV, I loaded a color image and inspected its shape to identify its height, width, and color channels, learning that OpenCV defaults to a BGR color ordering rather than RGB. By converting the image to grayscale, the three color channels were reduced into a single intensity channel. I also performed basic manipulations, such as cropping a region of interest (ROI), resizing the image dimensions, and rotating it by 90 degrees. These steps showed practically how modifying raw pixel data alters the visual output.

Lab 2: Edge Detection

Based on the edge detection script, the Sobel and Canny methods produce very different edge maps. The Sobel method detects edges by calculating gradients along the x and y directions. Because it relies on basic gradient magnitude, its resulting edge map usually looks thicker and captures broader changes in intensity. Canny edge detection, however, uses gradients combined with thresholding to specifically target strong edges. As a result, the Canny edge map is much cleaner, producing thin, sharp white lines on a black background and effectively filtering out the extra noise that Sobel tends to pick up.

Conclusion:

In this lab activity, I learned how to implement fundamental edge detection algorithms to extract structural features from an image. By comparing the results side by side, I saw that the Sobel method is good for detecting broad gradient changes along the x and y axes, but the Canny method produces much cleaner and sharper outlines because it uses thresholding to target strong edges. Overall, this activity showed me how important preprocessing and edge extraction are as foundational steps for higher-level computer vision tasks like object detection and segmentation.

Lab 3: Histogram Equalization

1. What is histogram equalization?

   Answer: It is an image processing technique used to redistribute pixel intensity values over the entire range of the image.
2. Why do we need histogram equalization?

   Answer: We need it to improve the overall contrast of an image, which allows for better visibility of details in images that suffer from poor lighting or low contrast.
3. How does histogram equalization work?

   Answer: It works by taking the image's histogram and spreading out the most frequent intensity values, effectively stretching the contrast across the full available spectrum.
4. How does CLAHE work?

   Answer: Contrast Limited Adaptive Histogram Equalization (CLAHE) works by applying equalization adaptively rather than globally. It divides the image into small local regions (tiles) and equalizes them individually, which prevents the over-amplification of noise that happens with standard global equalization.
5. What is the possible real-world application for histogram equalization?

   Answer:A common application is in medical imaging, like enhancing X-rays or MRI scans. Doctors need to improve the contrast to see subtle fractures or tissue differences clearly without washing out the rest of the scan with noise.

Conclusion:

Through this activity, I gained hands-on experience using histogram equalization to improve the contrast and visibility of details in poorly lit images. I observed that while standard global equalization helps stretch the pixel intensities over the entire range, it can sometimes wash out the image or amplify artifacts. Applying Contrast Limited Adaptive Histogram Equalization (CLAHE) provided a much better result because it adjusts the contrast locally without over-amplifying noise. This proved how vital contrast adjustment is for preparing images for further analysis.

Lab 4: SURF Feature Detection

1. What effect does changing the Hessian threshold have on the number of detected keypoints?

   Answer: Changing the Hessian threshold controls the sensitivity of the detector. A higher threshold results in fewer keypoints being detected because the algorithm will only select the most prominent, strong features in the image.

2. What are the descriptors in SURF used for, and why are they important?

   Answer: Descriptors mathematically describe the area around a detected keypoint. They are important because they allow these features to be matched between different images, even if the image is scaled or viewed differently.

3. How does SURF compare to SIFT and ORB in feature detection and speed?

   Answer: SURF is optimized to be faster than SIFT while still detecting robust, blob-like features using integral images. However, both SURF and SIFT are patented algorithms, meaning they may require licensing for commercial use. ORB is a free, unpatented alternative that can be used for feature detection when SURF isn't available.

Lab 5: PCA in Computer Vision

1. Why is dimensionality reduction important in computer vision?

   Answer: Raw image data is highly dimensional, which causes the "curse of dimensionality". Reducing dimensions makes machine learning algorithms less computationally expensive and helps prevent overfitting.

2. What is the difference between the first principal component and the second principal component?

   Answer: The first principal component captures the single direction of maximum variance in the dataset. The second component captures the next highest variance that is completely uncorrelated (orthogonal) to the first.

3. If PCA reduces an image from 1,000 features to 100 components, what information might be lost?

   Answer: If PCA reduces an image from 1,000 features to 100 components, it loses high-frequency noise, minor background details, and subtle pixel variations, retaining only the most significant structural features like the shape of the face.

4. How can PCA be applied to face recognition?

   Answer: PCA extracts the most significant statistical variations among faces to create "Eigenfaces". Any human face can be approximated as a weighted linear combination of these Eigenfaces, allowing models like an SVM to classify and recognize individuals using just these few PCA weights instead of thousands of raw pixels.

5. Why should image features generally be centered before applying PCA?

   Answer: Image features must be centered before PCA so that the algorithm accurately calculates the variance based on differences between the images, rather than being skewed by the overall average brightness of the dataset.

Lab 6: YOLOv11 Object Detection

Object detection is a computer vision task that finds specific objects in an image and draws bounding boxes around them to show their exact location, often predicting labels and confidences in real-time. This differs from image classification, which only assigns a single label to the entire image without providing any location data. For example, in a self-driving car system, image classification would simply analyze a camera feed and state "there is a pedestrian in this image." Object detection, however, will draw precise bounding boxes around the pedestrian, a stop sign, and other cars in real-time. This location data is exactly what allows the car's computer to know where the person is so it can apply the brakes.

Lab 7: Segmentation Techniques

Thresholding Segmentation: Segments an image into foreground and background by checking if pixel values meet a specific threshold.

- Advantage: It is best for simple, high-contrast images and is very fast to compute.
- Limitation: It struggles heavily with non-uniform illumination or shadows.

Region-Based Segmentation: Groups pixels based on similarity criteria and uses seed points to grow regions, such as the watershed algorithm.

- Advantage: It is highly effective at separating objects that are touching each other.
- Limitation: It faces major challenges with seed selection and often leads to over-segmentation.

Edge-Based Segmentation: Uses operators like Canny to highlight object boundaries and uses those edges to detect closed contours.

- Advantage: Great for capturing the structural shape and boundaries of an object.
- Limitation: It is highly sensitive to noise and relies heavily on having clear edges.

Semantic Segmentation: Uses pre-trained deep learning models (like DeepLab or U-Net) to label every single pixel in an image with specific class information.

- Advantage: Provides highly detailed scene understanding for complex datasets.
- Limitation: It has high computational complexity and requires deep learning frameworks to run.

Conclusion:

This laboratory exercise helped me understand how to separate and identify objects within an image using various segmentation methods. I found that simple thresholding is fast but struggles with uneven lighting, while region-based methods like watershed are great for separating touching objects but can easily lead to over-segmentation. Meanwhile, edge-based segmentation captures boundaries well but is highly sensitive to noise. Finally, applying semantic segmentation with a pre-trained deep learning model showed me how powerful AI is for detailed scene understanding, though I realized it comes at the cost of much higher computational complexity.